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EXECUTIVE SUMMARY OF THE THESIS

A Data-Driven Multi-Armed Bandit Framework for Adaptive Game-Based Handwriting Training in First-Grade Students

LAUREA MAGISTRALE IN BIOMEDICAL ENGINEERING - INGEGNERIA BIOMEDICA

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1. Introduction

Handwriting represents an essential educational skill mastered by third grade. Some children can struggle with it, either due to a Specific Learning Disorder (SLD), namely dysgraphia, or due to a temporary delay. Difficulties can manifest in different ways and often go unnoticed, leading children not only to fail to meet school expectations but also to feel frustrated and low in self-esteem. This is due to the absence of standardized diagnostic criteria and objective data, combined with limited resources for early detection [1]. Since difficulties can be variable and unrecognized until handwriting is mastered, technology can be a powerful tool: it can support students, provide engaging, eventually personalized training activities, and reduce stress on mental health [4]. In particular, the use of serious games can play a key role in maintaining high motivation, allowing children to improve through fun. Serious games are video games designed for other purposes beyond entertainment, such as academic learning. On the other hand, the effectiveness of a serious game is linked to player adherence and engagement [3], but sustaining engagement is no small challenge. To this end, several strategies have been proposed;

among them, adaptive approaches can tailor the game to players' behavior and improve learning outcomes [6].

1.1. Related works

In recent research, machine learning techniques, such as reinforcement learning (RL) and its Multi-Armed Bandit (MAB) variant, have been implemented, although their potential remains underexplored in educational contexts. Notably, RL and MAB, which exist in both stationary and dynamic variants, are already established in gaming domains, such as Dynamic Difficulty Adjustment (DDA) frameworks [2] and personalized game experiences [7]. Careful design could leverage RL and MAB techniques to sustain engagement in educational contexts as well, where a system is needed that automatically adapts to individual practice needs and provides direct control over the level of involvement.

1.2. Objectives

Given these premises, this work aims to develop a data-driven RL-based system to automatically provide difficulty progression in a game-based handwriting training. To achieve this primary objective, the following goals are pursued se-

quentially:

1. identifying the best MAB algorithm and configuration to best model the problem;
2. extending the analysis to a dynamic formulation to model evolving skill levels;
3. designing the deployment of the system in a real-world educational context and outlining a protocol for a future longitudinal study to validate its effectiveness in clinical and school settings.

2. Material and Methods

2.1. Number Tracing

Number Tracing is a handwriting video game developed in Unity and deployed on a 10.6-inch Samsung S6 Lite tablet equipped with an S-Pen. The game, developed before this study, was co-designed with neuropsychiatrists to help children develop fine motor control and enhance the precision needed for handwriting, as it requires tracing a number following the correct trajectory (Figure 1).



Figure 1: Screen of the game in which there is a number to trace.

It is possible to choose between three different kinds of difficulty:

1. **Easy**: there is a standard path width.
2. **Medium**: the path width is unchanged, but before tracing has to choose between a correct and a mirrored version of the number.
3. **Hard**: the path width is reduced.

2.2. Participants and Protocol

The study involved first-grade students from four classes across two primary schools in Varese province. Students had to participate in a 20-minute training session twice a week from Febru-

ary 17th, 2025, to May 5th, 2025. Each activity was explained by the researchers, who remained available for any questions. The protocol was approved by Politecnico di Milano’s Ethical Committee n. 52/2024.

2.3. Game Features Extraction

Previous work processed session-level raw signals (pressure, tilt, azimuth, speed) recorded by the S-Pen and extracted 21 features across 4 domains: kinematic, temporal, movement quality, and frequency. The feature space was completed by integrating game-specific performance features: accuracy, computed as the percentage of the trace within the boundaries, and stroke ratio, computed as the ratio of performed strokes to the expected number of strokes.

2.4. Static Modeling of the Problem

The aim of this work is to maximize the performance of a serious game over time, sustaining a high level of engagement. Hence, Number Tracing can be modeled as a constrained MAB problem, similar to what was developed by Pacchiano et al. [5]. To define the model, game elements were mapped onto the MAB framework.

- **Actions (Arms)**: the agent action set consists of three difficulty levels (easy, medium, and hard).
- **Reward**: is defined as a performance metric and should be a feature that reflects how effectively the player completes a given level.
- **Constraint**: to maintain player motivation, an engagement measure, selected from game features, was defined as a constraint. Therefore, the agent must ensure that an arm respects the imposed constraint.
- **Policy**: is the action-selection strategy. To test various strategies, ϵ -greedy (EG), Upper Confidence Bound (UCB), and Thompson Sampling (TS) algorithms were used.
- **Players**: were modeled as static players to evaluate algorithms under controlled and fixed data distributions.

2.4.1 General Framework

MAB is an online sequential decision-making problem where a learner pulls an arm $a_t \in \mathcal{A}$ at each round t . Then, the environment gives it a stochastic reward $r_t \in \mathbb{R}$ for only a_t . The

main objective is to maximize the expected cumulative reward over the time horizon T :

$$\max_{\pi} \mathbb{E} \left[\sum_{t=1}^T r_t \right], \quad (1)$$

where π is the strategy/policy used. Equation (1) is equivalent to minimizing cumulative regret, the loss caused by the selection of a suboptimal arm. Hence, the algorithm has to balance between exploration of new arms and exploitation of those already known.

In this work, a constrained version based on two features of the serious game is considered. **Accuracy**, which expresses how well a specific level was executed, was used as a reward. **Time-in-air** was selected as engagement measurement $e_t = T_{in\ air_t}$. It was assumed that a longer time passed with the pen lifted indicates boredom, so lower values are preferable. Then, a subset $\tilde{\mathcal{A}}_{feas} \subseteq \mathcal{A}$, including only feasible arms, was defined: an arm is considered feasible when its estimated value of time-in-air, computed according to MAB policies, is below the threshold θ , set on the 75th percentile of time-in-air distribution. If $\tilde{\mathcal{A}}_{feas} = \emptyset$, arms with the minimum time-in-air are treated as feasible.

The general pseudo-code is presented in Algorithm 1. After a three-step exploration phase ($|\mathcal{A}| = 3$), the learner evaluates if $eng_{i,t-1} \leq \theta$, where eng depends on the algorithm and represents an amount of time-in-air. This inequality defines $\tilde{\mathcal{A}}_{feas}$, and from it the learner selects the arm a_i according to its policy π based on $acc_{i,t-1}$. Once the stochastic reward $r_{a_t,t}$ and engagement value $e_{a_t,t}$ are observed, the statistics of a_t are updated using the MAB method. If $\tilde{\mathcal{A}}_{feas} = \emptyset$, the learner will select a random arm from $\mathcal{A}$, to allow exploration.

To evaluate algorithm performance, real averages of the data distribution are used. A subset $\mathcal{A}_{feas} \subseteq \mathcal{A}$ was defined as $\mathcal{A}_{feas} = \{a_i \in \mathcal{A} \mid \mu_e(a_i) \leq \theta\}$, or, if all arms exceed the threshold, $\mathcal{A}_{feas} = \arg \min_{a \in \mathcal{A}} \mu_e(a)$. This second definition prioritizes engagement over performance. From the $\mathcal{A}_{feas}$ subset, the best arm was calculated as $\mu^* = \max_{a \in \mathcal{A}_{feas}} \mu_a(a)$. If the agent selects an arm $a \notin \mathcal{A}_{feas}$, in the regret calculation (Equation (2)), a penalty of 0 was applied.

$$\begin{cases} R = T \cdot \mu^* - \mathbb{E} \left[\sum_{t=1}^T r_t \right], & a_t \in \mathcal{A}_{feas} \\ R = T \cdot \mu^* - 0, & a_t \notin \mathcal{A}_{feas} \end{cases} \quad (2)$$

Algorithm 1 Framework

Require: Input: $\mathcal{A}, \theta, T$

```

1: Initialize  $acc_{i,t=0}, eng_{i,t=0} \leftarrow 0$ 
2: for each  $a_i \in \mathcal{A}$  do
3:    $a_t \leftarrow a_i$ 
4:   Observe  $r_{a_t,t}, e_{a_t,t}$ 
5:   Update statistics of arm  $a_t$ 
6:    $t \leftarrow t + 1$ 
7: end for
8: for  $t = |\mathcal{A}| + 1$  to  $T$  do
9:    $\tilde{\mathcal{A}}_{feas} \leftarrow \{a_i \in \mathcal{A} \mid eng_{i,t-1} \leq \theta\}$ 
10:  if  $\tilde{\mathcal{A}}_{feas} = \emptyset$  then
11:    Choose  $a_t \in \mathcal{A}$ 
12:  else
13:    Choose  $a_t \in \tilde{\mathcal{A}}_{feas}$  according to  $\pi$  based
      on  $acc_{i,t-1}$ 
14:  end if
15:  Observe  $r_{a_t,t}, e_{a_t,t}$ 
16:  Update statistics of arm  $a_t$ 
17: end for

```

2.4.2 Selection of the Best Algorithm

Three algorithms were compared, each implementing a different action-selection strategy.

Epsilon-greedy (EG) With probability $1 - \epsilon$, the agent selects the arm with the highest estimated accuracy from $\tilde{\mathcal{A}}_{feas}$ (exploitation); otherwise, it picks a random arm from $\mathcal{A}$ (exploration). Two variants were considered: the first with $\epsilon = 0.1$ and the second with $\epsilon = 0.2$.

Upper Confidence Bound (UCB) This approach optimistically overestimates accuracy via UCB and underestimates time-in-air via Lower Confidence Bound (LCB), both computed as $\sqrt{2 \ln(t) / N_a}$. Feasible arms are filtered through LCB, and the one with the highest UCB is selected.

Thompson Sampling (TS) Each action is modeled by two beta distributions (accuracy and time-in-air), initialized as $\mathcal{B}(1, 1)$. At each step, feasible arms are identified by sampling from the time-in-air distributions, and the arm with the highest accuracy sample is selected. Parameters are updated after each observation.

2.4.3 Metrics Performance

For algorithm evaluation, several performance metrics have been considered.

- **Cumulative reward:** it is the sum over time of the reward observed by the learner. It is a direct measure of an algorithm’s effectiveness in maximizing rewards.
- **Mean reward:** measures the average reward collected over the time horizon T . Let to understand how much and how quickly an algorithm converges to μ^* .
- **Cumulative regret:** indicates the performance gap between what the agent selects and the optimal policy.

Since algorithms are intrinsically stochastic, multiple runs were performed to calculate 95% confidence intervals and better capture the range of possible outcomes and quantify run-to-run variability.

2.4.4 Selection of the Best Configuration

Three different operational configurations were identified to assess the algorithms’ performance.

1. **individual:** each player is considered in their uniqueness. The data available to MAB methods are child-specific.
2. **aggregated:** all data collected were used to perform a single simulation.
3. **cluster-based:** two clusters are derived from a K-Medoids clustering performed on accuracy and time-in-air features: proficient children, who have a high level of accuracy and short time-in-air, and struggling children, who have greater difficulty completing the trace and higher time-in-air.

2.4.5 Evaluation protocol

Each algorithm is tested in all settings in Python 3.10.2, with a time horizon $T = 10000$, and the experiment is repeated 10 times. All methods are analyzed by performing uniform random re-sampling of the specific filtered dataset. To determine the best algorithm-configuration pair, for each run both cumulative reward and regret were considered at the time step corresponding to the final real session. Since cumulative regret is penalized, cumulative reward is used as the main quantitative performance metric. For both individual- and cluster-based assets, all values

were summed. The mean and confidence intervals were calculated, allowing the best pair to be identified.

2.5. Dynamic Modeling of the Problem

Since the static model does not reflect real player improvements, moving to a dynamic model is the natural step forward. A chronological window-based approach was implemented, with each window containing at least 30 samples per arm. Given the non-homogeneous distribution of data between the three levels, the initial sessions were excluded. Every time the window changes, the set of feasible arms A_{feas} and the best arm have to be recalculated.

2.5.1 Selection of the Best Algorithm

In addition to the three algorithms used in the static approach, Linear UCB (LinUCB) was considered: it is suitable for the dynamic setting since it adapts its parameters to new observations.

LinUCB LinUCB extends the UCB approach to the contextual setting, incorporating user context $\mathbf{x}_t$ to estimate both expected reward and time-in-air. As in UCB, feasible arms are filtered via LCB, and the arm with the highest UCB is selected from $\tilde{\mathcal{A}}_{feas}$. The confidence bounds are scaled by a coefficient α , which controls exploration based on uncertainty and context novelty. The context $\mathbf{x}_t$ models learning trajectories via a constrained exponential plateau ($R^2 = 0.719$): $\mathbf{x}_t = 1 - e^{-t/\tau}$, with τ fitted on accuracy data.

2.5.2 Evaluation protocol

Also, in a dynamic setting, evaluation follows the protocol reported in Section 2.4.5, but only one configuration is considered.

2.6. Deployment

To evaluate the static framework in a real educational context, the best-performing algorithm can be deployed in the game using its final learned parameters. Since the knowledge acquired depends on the specific algorithm and its training phase, a well-trained model requires fewer additional data to achieve optimal performance in deployment. A future longitudi-

nal study could validate the system’s impact by comparing an experimental group playing the data-driven adaptive game against a control group using the standard version.

3. Results

A total of 66 first-grade students (31 females and 35 males, 6.45 ± 0.31 , 6 left-handed) participated in the study. Each child has played in the median 88 [62.5, 116.5] times, with the hard level as the most-played level (54% of total data).

3.1. Simulations

In each run, the average mean reward and cumulative regret over 10 runs were plotted. Algorithms were optimized based on data: from time-in-air distribution, θ threshold was selected as the 75th percentile, corresponding to $Q3 = 0.35235$, and the τ used in the LinUCB context was derived from the real trajectory of the accuracy, amounting to 21.

A total of 670 simulations were computed. In the aggregated and cluster-based configuration, the system behaved as a standard MAB, since all arms remained feasible throughout. Conversely, in the individual configuration, three distinct behaviors emerged: all arms are always feasible, one or two arms become unfeasible, or no feasible arm is available.

3.2. Best algorithms

In the static case, Aggregated-TS achieved the highest score in reward and the lowest score in regret (cumulative reward: 5348.487 ± 9.325 , cumulative regret: 22.567 ± 7.545), while LinUCB was the best-performing algorithm in the dynamic setting (cumulative reward: 2360.033 ± 4.544 , cumulative regret: 181.455 ± 24.287). In Table 1, all scores related to cumulative reward are reported.

3.3. Deployment

Since Aggregated-TS results as the best pair, its posterior distribution (Figures 2) can be used as new priors in a deployment phase.

4. Discussion

This study developed a constrained MAB framework for adaptive difficulty selection in Number Tracing, a handwriting training game for first-

Setting	Algorithm	CumReward
<i>Individual</i>	UCB	4879.274 ± 4.480
	TS	4628.369 ± 41.557
	EG $\epsilon = 0.1$	4511.050 ± 63.566
	EG $\epsilon = 0.2$	4490.201 ± 55.693
<i>Aggregated</i>	UCB	5287.151 ± 5.657
	TS	5348.487 ± 9.325
	EG $\epsilon = 0.1$	5253.090 ± 14.454
	EG $\epsilon = 0.2$	5144.632 ± 7.909
<i>Cluster</i>	UCB	5290.492 ± 4.019
	TS	5269.123 ± 32.196
	EG $\epsilon = 0.1$	5154.716 ± 64.146
	EG $\epsilon = 0.2$	5070.311 ± 32.896
<i>Dynamic</i>	UCB	2327.360 ± 3.216
	TS	2338.153 ± 13.769
	EG $\epsilon = 0.1$	2284.666 ± 12.769
	EG $\epsilon = 0.2$	2260.998 ± 13.074
	LinUCB	2360.033 ± 4.544

Table 1: Score of cumulative reward with the best results highlighted.

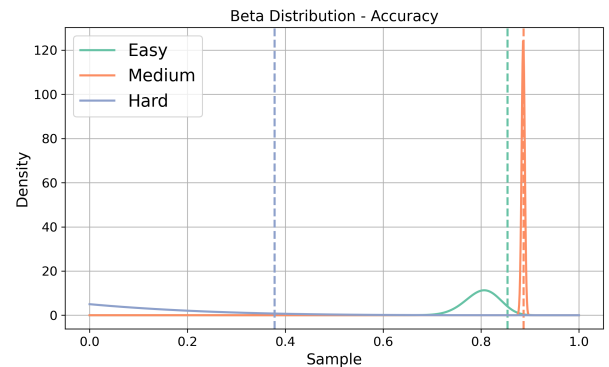


Figure 2: Aggregated-TS posterior distributions of accuracy.

grade students. Accuracy served as the reward signal, while time-in-air served as the engagement constraint.

In the static setting, TS consistently outperformed the other algorithms, converging faster to μ^* with the lowest cumulative regret. UCB exhibited performance similar to TS but converged more slowly to the optimal policy. From a variability perspective, UCB was the algorithm with the highest stability. Both EG variants showed the worst performance: their constant exploration rate prevents full exploitation, with EG ($\epsilon = 0.2$) yielding the largest gap to μ^* .

In the dynamic setting, LinUCB achieved the best performance in both mean reward and cu-

mulative regret, leveraging contextual information to explore more efficiently. The remaining algorithms followed the same relative ranking observed in the static case, with TS and UCB showing logarithmic regret growth that flattened at the last window transition, while both EG variants maintained linear trends.

The aggregated configuration outperformed individual and cluster-based settings across all algorithms. This advantage is primarily attributable to data availability rather than to a genuine superiority of the global approach: the larger dataset reduces variance in the estimates and ensures that all arms remain feasible throughout, effectively simplifying the problem to a standard unconstrained MAB. The cluster-based configuration, derived from a posterior analysis of accuracy and time-in-air mediated over the entire training, also outperformed the individual one. However, this information would not be available in real-time deployment, limiting its direct applicability.

4.1. Limitations and Future Work

Several limitations should be acknowledged. First, the relatively scarce number of sessions (median 88 per child) and the heterogeneous distribution across difficulty levels constrain the reliability of individual-level analyses; larger datasets are needed to properly evaluate the individual approach. Second, both the cluster assignments and the contextual feature $\mathbf{x}_t$ used in LinUCB were derived from data available only at the end of the training period. While this limits their direct use in online deployment, the estimated τ parameters could inform the design of future LinUCB implementations. Third, the current model does not account for skill transfer across difficulty levels: handwriting improvements may occur even when a specific arm is not played, and future formulations should incorporate cross-arm learning effects. However, the insights gathered during this preliminary analysis can inform a future study, which would particularly benefit from a randomized controlled trial design, in which some children use the MAB-based difficulty selector while others do not, allowing for a true evaluation of this system’s efficacy.

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